



SmartElex Digital Speaker Module



Key Features:

- **Compact audio output unit** — works as a sound generator in embedded systems.
- **Dual functionality** — straight speaker output + buzzer-like alerts.
- **Wide operating voltage range:** ~2.0 V to 5.5 V DC — flexible for 3.3 V (ESP32) or 5 V (Arduino).
- **Simple interface:** connect via a few pins (power, ground, signal).
- Built-in **potentiometer for volume control** (on some variants).
- Suitable for **sound effects, notifications, simple audio playback**.

Typical Connections:

Module Pin	Connects To	Function
VCC	3.3 V / 5 V (MCU power)	Power supply
GND	GND	Common ground
SIGNAL	Digital I/O	Audio input from MCU (PWM / tone)

Note: Exact pin names and layout depend on the module version — check the specific board labeling.

How It Works:

- The module typically has a **built-in driver/amplifier** and speaker element, so you can send digital audio signals (like PWM from Arduino) directly to it.
- For **simple beeps or tones**, you can use Arduino's `tone()` function; for ESP32, use PWM or DAC outputs.
- Not designed for high-quality music playback — best for **alerts, effects, voices**, simple audio.

Example: Arduino Tone Output:

```
#define SPEAKER_PIN 8
```

```
void setup() {  
  pinMode(SPEAKER_PIN, OUTPUT);  
}
```

```
void loop() {  
  tone(SPEAKER_PIN, 1000); // 1 kHz beep  
  delay(500);  
  noTone(SPEAKER_PIN);  
  delay(500);  
}
```

✓ Produces simple tones through the speaker module.

✓ Works when module SIGNAL pin is connected to the Arduino pin above.

Notes & Tips:

- ◆ This module **isn't a DAC audio decoder** — to play audio files (MP3/WAV) you'd use a player module (like DFPlayer Mini) feeding this speaker.
- ◆ For **ESP32 PWM or DAC output**, choose a pin that supports PWM/DAC (e.g., GPIO 25/26).
- ◆ Ensure power supply voltage matches the module's supported range (2–5.5 V).

Good For:

- ✓ Arduino/ESP32 notification sounds
- ✓ Beep alerts in user interfaces
- ✓ Simple voice playback (with a proper audio source)
- ✓ DIY sound effects for robots, toys, alarms